Support Vector Machines

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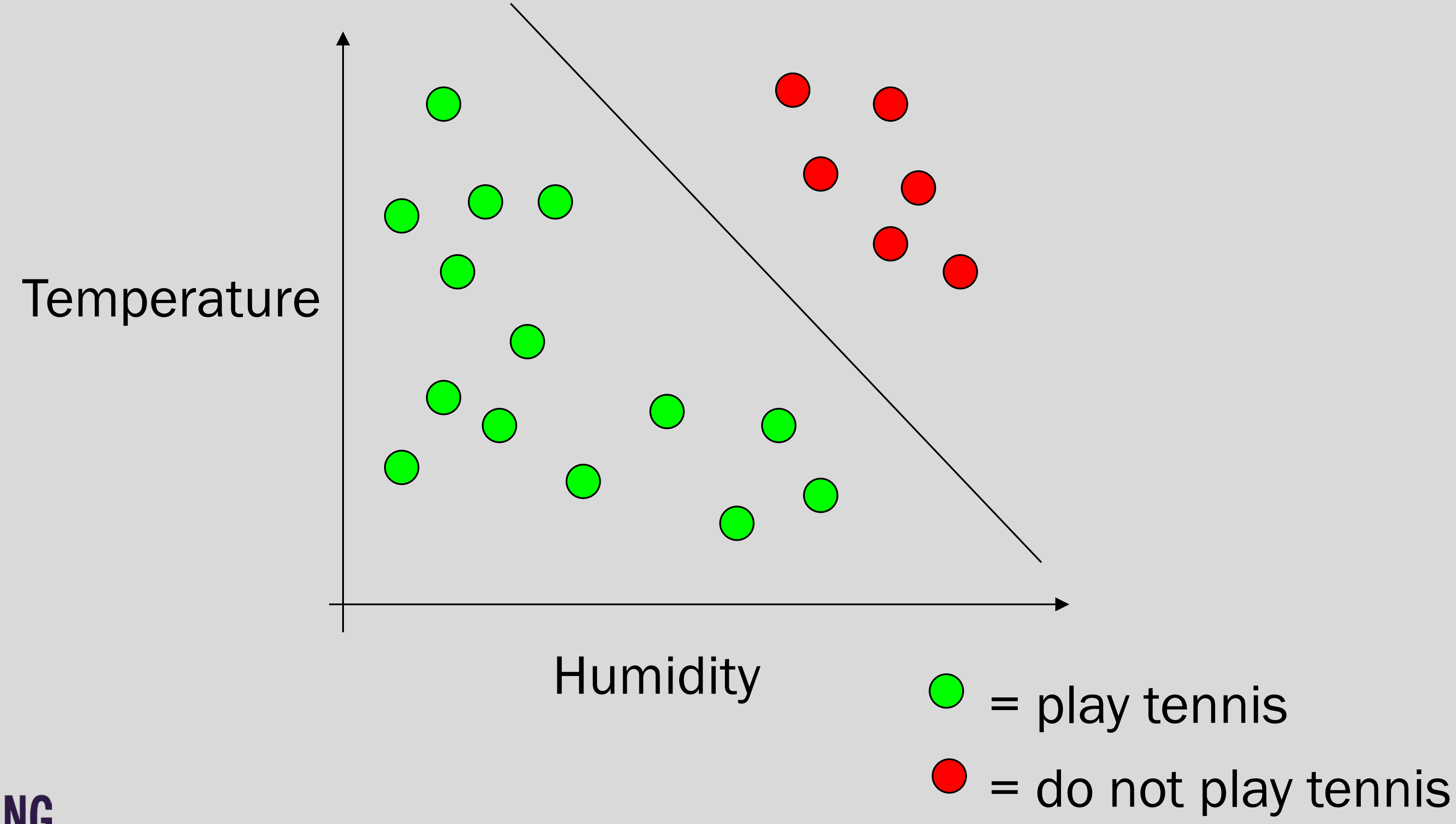
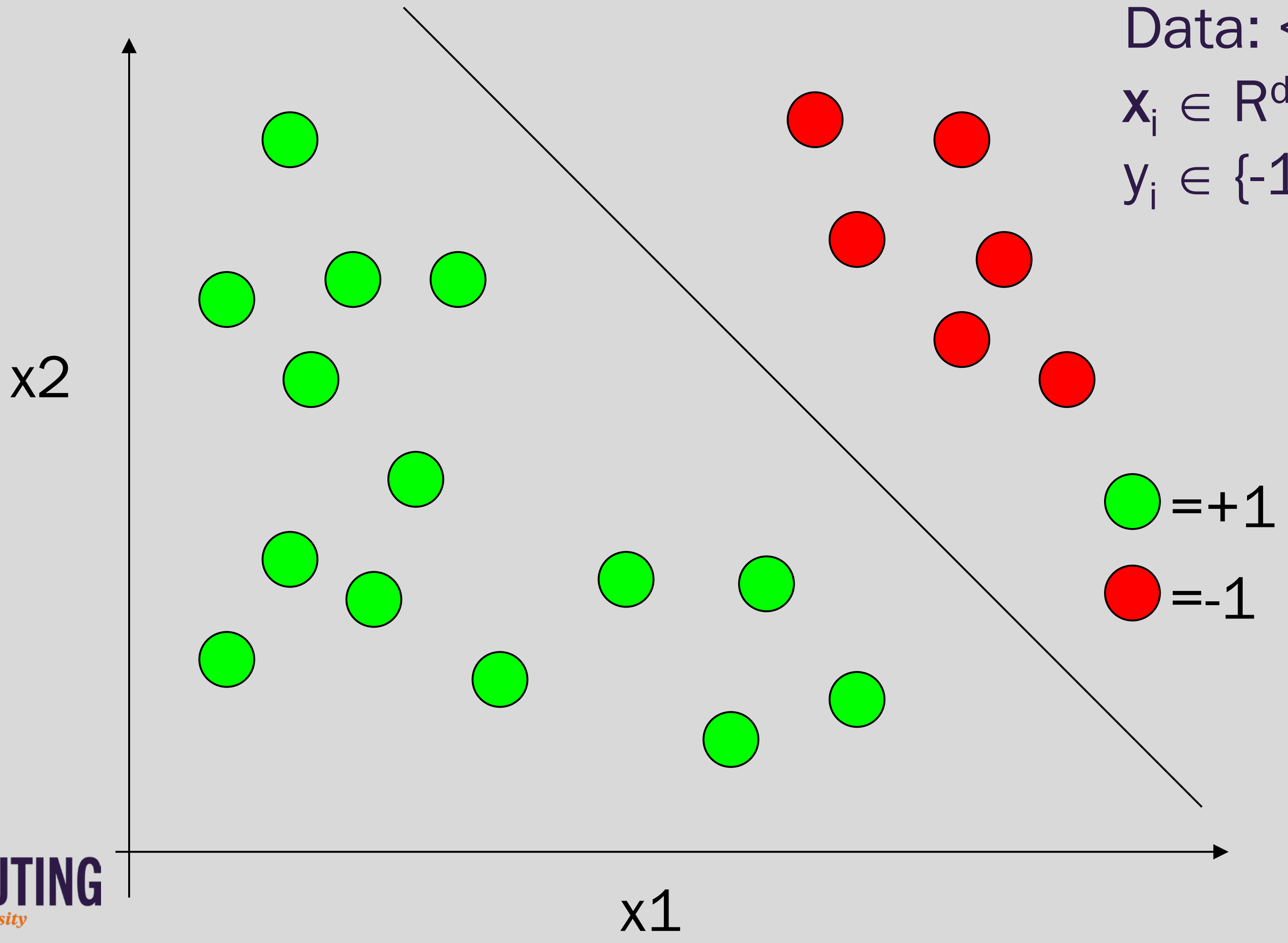
Main Ideas

- Max-Margin Classifier (Separating hyperplane)
  - Formalize notion of the best linear separator
- Lagrangian Multipliers
  - Way to convert a constrained optimization problem to one that is easier to solve
- Kernels
  - Projecting data into higher-dimensional space makes it linearly separable
- Complexity
  - Depends only on the number of training examples, not on dimensionality of the kernel space!

Tennis example



Linear Support Vector Machines



Data: $\langle \mathbf{x}_i, y_i \rangle$, $i=1,\ldots,l$

$\mathbf{x}_i \in R^d$

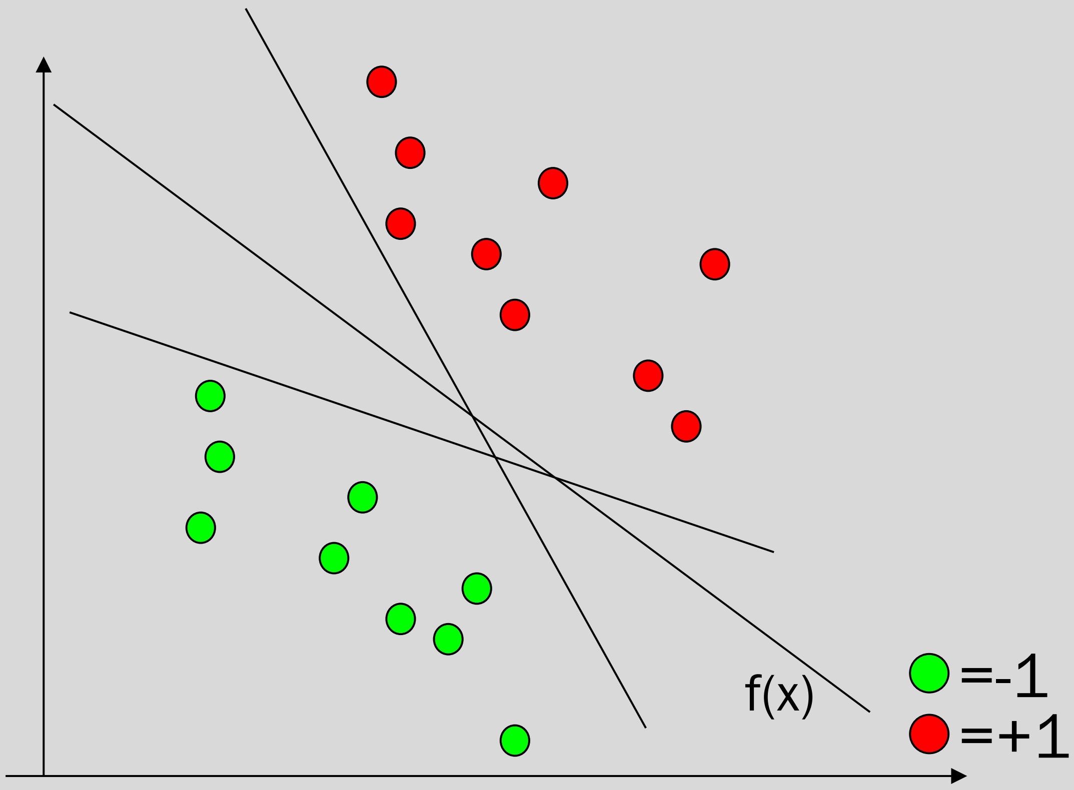
$y_i \in \{-1,+1\}$

Linear SVM 2

All hyperplanes in R^d are parameterize by a vector ($\mathbf{w}$) and a constant b.

Can be expressed as $\boldsymbol{w} \bullet \boldsymbol{x} + b = 0$ (remember the equation for a hyperplane from algebra!)

Our aim is to find such a hyperplane <u>f(x)=sign(**w**•**x**+b),</u> that correctly classifies our data.



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